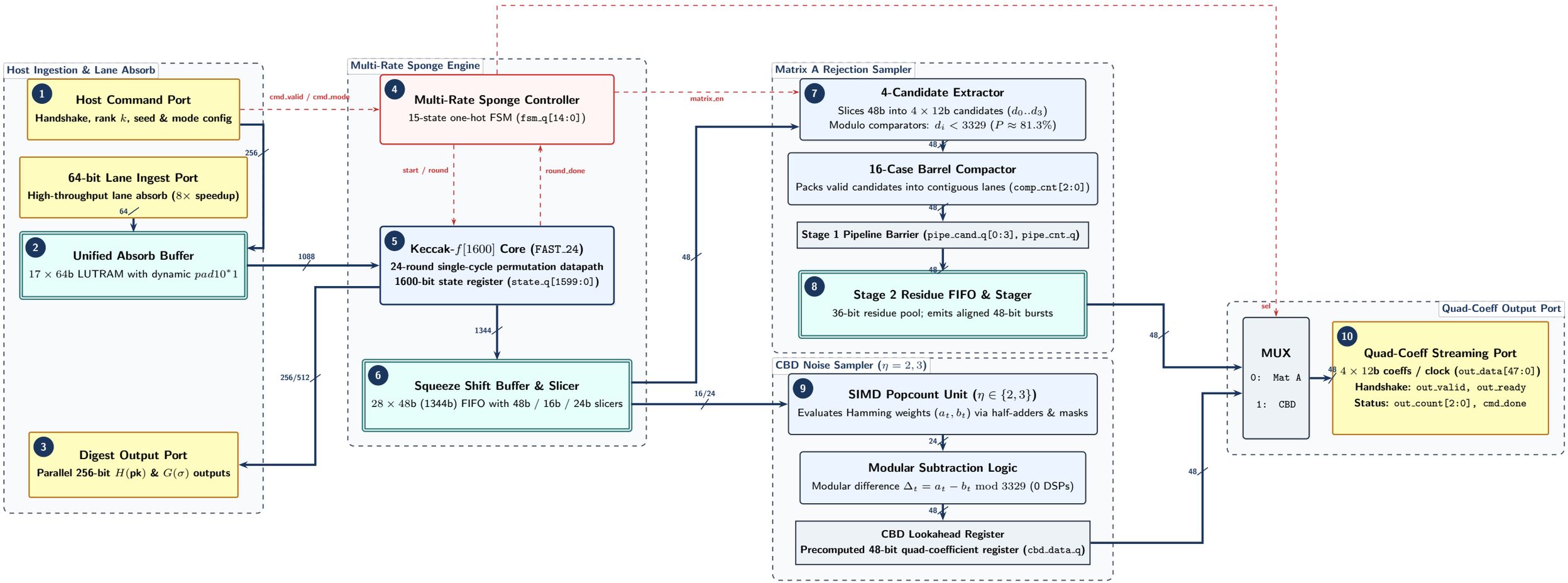
ML-KEM HPKA: Unified Cryptographic Hash Sponge, Matrix Sampler & CBD Unit

Subsystem 09 Microarchitecture — RTL Modules: `mlkem_hash_sampler.sv`, `keccak_f1600.sv`, `sha3_256_core.sv`



ID	Component / Signal	Width / Format	Description
1	Host Command Port	Handshake / Config	Latches mode, rank k, seeds, and swaps coordinates $(i, j) \leftrightarrow (j, i)$ for zero-cycle matrix transposition.
2	Fast 64-bit Absorb Buffer	17×64b (1088-bit) LUTRAM	Absorbs 64-bit lanes per cycle ($8\times$ byte speedup) and injects dynamic $pad10^*1$ padding at lane boundaries.
3	Parallel Digest Output Port	2×256-bit Hash Buses	Emits single-cycle latched SHA3-256 H(pk) and upper 256-bit $G(\sigma)$ digest outputs to host registers.
4	Multi-Rate Sponge Controller	15-bit One-Hot Register	Orchestrates absorption, permutations, and squeezing across SHA3-256/512 and SHAKE-128/256 modes.
5	Keccak-$f[1600]$ Core (FAST_24)	1600-bit State Datapath	Executes 24-round permutation in 24 clock cycles; provides single-cycle θ-ρ-π-χ-ι round logic.
6	Squeeze Shift Buffer & Slicer	28×48b (1344-bit) FIFO	Streams 48b words for Matrix A or 16b/24b for CBD, stitching byte 135 across blocks for $\eta = 3$ sampling.
7	4-Way Rejection Candidate Engine	4×12-bit Modulo Datapath	Extracts four 12-bit candidates ($d_0..d_3$), tests $d_i < 3329$ ($P \approx 81.3\%$), and compacts items in 1 LUT delay.
8	Stage 2 Residue FIFO & Stager	36-bit Residue + 48-bit Staging	Buffers leftovers and emits strictly aligned 4-coefficient bursts (48 bits), shielding NTT cores from stalls.
9	CBD SIMD Noise Sampler	SIMD Popcount + 6-LUT ALU	Computes Hamming weights for $\eta \in \{2, 3\}$ and evaluates modular difference $a - b \bmod 3329$ with zero DSPs.
10	Quad-Coeff Streaming Port	48-bit Stream (4×12b / cc)	Multiplexes Matrix A and CBD streams into aligned 48-bit beats with ready/valid backpressure handshaking.